Market Basket Analysis for Understanding Customer Purchase Behavior

A Mid term report for the BDM capstone Project

Submitted by

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1. Executive Summary

This project, “Market Basket Analysis for Understanding Customer Purchase Behavior,” was conducted using three months of transactional sales data from Nana Grocery Store, a neighborhood retail outlet in Balasore, Odisha. The objective of the study was to analyze customer purchasing patterns, identify frequently co-purchased products, and derive actionable insights to support data-driven retail decision-making.

A structured analytical workflow combining Exploratory Data Analysis (EDA) and Market Basket Analysis (MBA) was implemented. The EDA phase summarized and visualized category-wise performance, sales trends, and customer spending distribution. This was followed by MBA using the Apriori algorithm, which identified statistically significant product associations based on support, confidence, and lift metrics.

Key findings revealed that Staples and Beverages are the highest revenue-generating categories, collectively accounting for over 40% of total sales. Products such as Aashirvaad Atta (5kg), Boost 500g, Surf Excel 1kg, and Horlicks 500g emerged as the top contributors to overall revenue, confirming strong household preference for essential grocery and branded FMCG items.
The average bill value ranged between ₹250 and ₹300, showing that most customers make small, frequent purchases consistent with local, need-based buying behavior.

Market Basket Analysis further uncovered high-lift product pairs, including (Eggs – Potato – Wheat Rava) and (Rice Bran Oil – Everest Red Chilli), which reflect complementary purchasing habits around meal preparation and daily cooking essentials. These associations present clear opportunities for cross-promotions, combo discounts, and optimized shelf placement strategies.

Overall, the analysis demonstrates that even small retail stores can leverage basic data analytics to enhance business efficiency. By applying evidence-driven insights to product bundling, inventory management, and in-store marketing, Nana Grocery Store can increase average transaction value, strengthen customer retention, and transition from intuition-based to data-informed retail operations.

2. Proof of originality of the Data

Proof of Originality of Data As part of the proof of originality for the collected data, I am providing the following supporting documents and materials: Letter from the Organization, Images and a video capturing an interaction with the founder. These are securely attached via a Google Drive link for your reference and review:

- BDM

(copy and paste the link in browser please) The mode of the video is in Odia Language majorly and an English transcript is provided in the video for your reference and review.. Photos of the shop have been added in the same drive link for your reference.

3. Metadata and Descriptive Statistics

3.1 Data Source and Collection Process

The dataset was collected from Nana Grocery Store, a neighborhood retail outlet in Kupari Market, Balasore. The store provided access to three months of digital sales data (August 1 – October 31, 2025) recorded from its daily billing and QR payment logs. Data was verified with the store owner, cleaned in Excel, and standardized to correct product naming inconsistencies and remove duplicates. The finalized dataset includes product details, transaction records, and summarized sales metrics, forming a structured foundation for Market Basket Analysis (MBA).

3.2 Metadata Description

The dataset comprises three interrelated datasets — Product Data, Transaction Data, and Sales Summary Data — each serving a specific analytical purpose. The metadata table below explains all variables, their data types, and their relevance to the project.

Dataset	Variable	Description	Data Type	Relevance / Purpose
Product Data	Product_ID	Unique alphanumeric identifier for each product	Categorical	Ensures each item is uniquely referenced and avoids duplication
Product Data	Product_Name	Name of the product sold	Text	Used to link with transaction records for analysis
Product Data	Category	Type or class of product (e.g., Staples, Beverages)	Categorical	Enables segmentation and category-wise trend analysis
Product Data	Unit_Price	Price per unit of the product	Numerical	Used to calculate total sales value and compare price ranges
Dataset	**Variable**	**Description**	**Data Type**	**Relevance / Purpose**
Transaction Data	Transaction_ID	Unique identifier for each bill	Categorical	Defines each shopping basket for Market Basket Analysis
Transaction Data	Date	Date of purchase	Date	Helps identify daily or seasonal purchasing trends
Transaction Data	Time	Time of purchase	Time	Used to identify customer buying patterns throughout the day
Transaction Data	Product_Name	Item purchased within the transaction	Text	Links to product master data for consistency
Transaction Data	Category	Product's classification	Categorical	Allows cross-category association analysis

Transaction Data	Quantity	Units of each item purchased	Numerical	Measures demand and bulk buying frequency
Transaction Data	Unit_Price	Selling price of the product	Numerical	Validates consistency with master product list
Transaction Data	Total_Price	Total amount for that item (Quantity × Unit_Price)	Numerical	Basis for computing total bill value and revenue
Dataset	**Variable**	**Description**	**Data Type**	**Relevance / Purpose**
Sales Summary Data	Product_Name	Product identifier	Text	Used to aggregate performance at the product level
Sales Summary Data	Category	Product's category	Categorical	Enables comparison of performance across categories
Sales Summary Data	Total_Quantity_Sold	Total units sold for each product	Numerical	Highlights fast-moving and slow-moving products
Sales Summary Data	Total_Sales_Value	Total revenue generated per product	Numerical	Identifies top-performing products
Sales Summary Data	No_of_Transactions	Number of bills containing the product	Numerical	Measures product popularity
Sales Summary Data	Avg_Transaction_Value	Average sales value per transaction for the product	Numerical	Reflects typical customer spending on each item

3.3 Summary of Dataset Characteristics

- Total Transaction Entries: 8,067 – Represents total product-level sales across three months, forming the basis for purchase pattern analysis.
- Unique Transactions (Bills): 2,482 – Indicates the total number of distinct customer purchase events.
- Unique Products: 82 items under 10 categories – Shows product diversity across segments like Staples, Snacks, and Beverages.
- Average Quantity Sold per Entry: 1.05 units – Suggests that most customers buy single units per item.
- Average Unit Price: ₹71.77 – Reflects a balanced mix of affordable and premium products.
- Average Total Price per Entry: ₹75.84 – Represents average revenue generated per item sold.
- Estimated Total Sales Value: ₹6.12 lakh (approx.) – Aggregated sales revenue over three months.
- Sales Duration: August 1 – October 31, 2025 – Continuous 92-day dataset representing both weekday and weekend patterns.

This structured dataset provides sufficient depth and breadth to analyze both product-level performance and customer purchasing behavior.

3.4 Descriptive Statistics

Descriptive statistics were computed for key numerical variables across all three data sheets to summarize sales distribution and pricing patterns.

Product Data: The mean unit price is ₹265, with a median and mode of ₹145, and a standard deviation of ₹318.5. Prices range from ₹22 to ₹1450, reflecting a wide variety of products. Most items fall in the mid-range, with a few premium goods contributing to the high price spread. This variability supports segmentation and cross-price association analysis.

Transaction Data: The average quantity sold per entry is 1.05 units, with little variation (SD = 0.22), indicating low-volume purchases. Unit prices average ₹71.77, while total prices per line item average ₹75.84, with a standard deviation of ₹88.28. The data shows that customers purchase few items per visit, but overall spending differs by price, enabling meaningful co-purchase pattern discovery.

Sales Summary Data: At an aggregated level, the mean total quantity sold per product is 104 units (SD = 46), and the mean total sales value is ₹7,462 (SD = ₹4,766). The average transaction count per product is 98, while the mean average transaction value is ₹128 (SD = ₹210). This high dispersion confirms that a few products contribute disproportionately to revenue — typical of small-scale retail behavior dominated by fast-moving staples.

These statistics provide a quantitative overview of the store's operations and directly support the project's objective. The combination of low average purchase quantities, high product diversity, and large variation in price and transaction value makes the dataset ideal for Market Basket Analysis. It reflects realistic customer purchasing behavior — small basket sizes but wide product variety — enabling the discovery of frequent itemsets and cross-selling opportunities. The observed dispersion in sales values indicates that certain products dominate the revenue mix, highlighting their importance in product bundling and shelf-placement recommendations.

4. Detailed Explanation of Analysis Process / Method

4.1 Analytical Framework

The analytical process adopted for this project combines **Exploratory Data Analysis (EDA)** and **Market Basket Analysis (MBA)** to extract insights from transactional data collected from Nana Grocery Store. The EDA phase focuses on summarizing, cleaning, and visualizing sales records to understand category contributions, sales frequency, and time-based variations.

The subsequent MBA phase applies the **Apriori algorithm** to identify products that are frequently purchased together and to quantify the strength of their associations. This two-phase workflow ensures both descriptive understanding and predictive pattern recognition. The entire approach is data-driven, leveraging a three-month dataset containing 8,067 product-level transactions.

4.2 Data Cleaning and Preprocessing

Before any analysis, a thorough data cleaning and preprocessing process was performed to ensure accuracy and consistency. The raw sales data extracted from the store's digital registers contained repeated product names, missing entries, and inconsistent category naming conventions. These issues were systematically resolved using **Microsoft Excel** and **Python (pandas)**.

Product names were standardized to match the *Product Data* master sheet, and categories were aligned to maintain uniform grouping. Duplicate rows were removed by identifying identical Transaction IDs and timestamps. Outlier detection was carried out in numeric fields such as *Unit Price* and *Total Price* using statistical thresholds (values beyond ±3 standard deviations were flagged for verification). In cases of mismatch, *Total Price* was recalculated as:

Total Price = Quantity × Unit Price

Additionally, the *Date* and *Time* fields were reformatted into a consistent datetime structure, ensuring the data could support time-series visualization. After preprocessing, a verified dataset of 8,067 clean and unique records was finalized for analysis.

This step was essential for improving data reliability, eliminating redundancy, and establishing a foundation for accurate association rule mining.

4.3 Exploratory Data Analysis (EDA)

The EDA phase provided an initial understanding of sales trends, customer buying habits, and category-level performance. Using **Excel Pivot Tables** and **Python visualizations (matplotlib, pandas)**, the data was explored across three key dimensions: category-wise sales, product performance, and temporal trends.

Key observations revealed that sales were evenly distributed through weekdays, with moderate increases during weekends. The *Staples* and *Beverages* categories consistently generated the highest revenue, while *Snacks* and *Personal Care* showed higher transaction frequency but lower per-item value.

EDA also analyzed time-of-day sales behavior, identifying evening hours as the busiest shopping window. These findings guided parameter selection for the Apriori algorithm by identifying

which products and categories to prioritize and setting minimum thresholds for support and confidence.

4.4 Market Basket Analysis (MBA)

After exploratory analysis, **Market Basket Analysis** was conducted using the **Apriori algorithm** implemented through the *mlxtend* Python library. The objective was to detect frequent itemsets and derive association rules that reveal products commonly purchased together.
Three key statistical measures were used to interpret these associations:

Support = (Number of transactions containing both A and B) ÷ (Total number of transactions)

Confidence = (Number of transactions containing both A and B) ÷ (Number of transactions containing A)

Lift = Confidence ÷ (Expected Confidence if A and B were independent)

Rules with **Lift > 1.2** were considered strong associations. After iterative refinement, the final parameters were set at **minimum support = 0.02** and **minimum confidence = 0.4**, ensuring the extracted rules were both statistically significant and practically meaningful.

Patterns observed included frequent co-purchases between *Rice* and *Sunflower Oil*, and between *Detergents* and *Soaps*, suggesting household-based purchase behavior. These insights have direct applications for cross-promotions and store shelf optimization.

4.5 Visualization and Interpretation

Visual tools were used to translate numerical findings into actionable insights.

- **Bar charts** were developed to visualize top-performing categories and products by total sales.
- **Line graphs** highlighted daily sales trends, revealing weekly cycles and consistent weekend demand spikes.
- **Network graphs** displayed product associations derived from Apriori rules, with node size indicating frequency and edge thickness representing lift values.

These visualizations clarified both sales concentration and associative strength among products. They not only identified high-performing items but also exposed interdependencies between categories, providing valuable business intelligence for inventory and marketing strategies.

4.6 Rationale for Chosen Analytical Approach

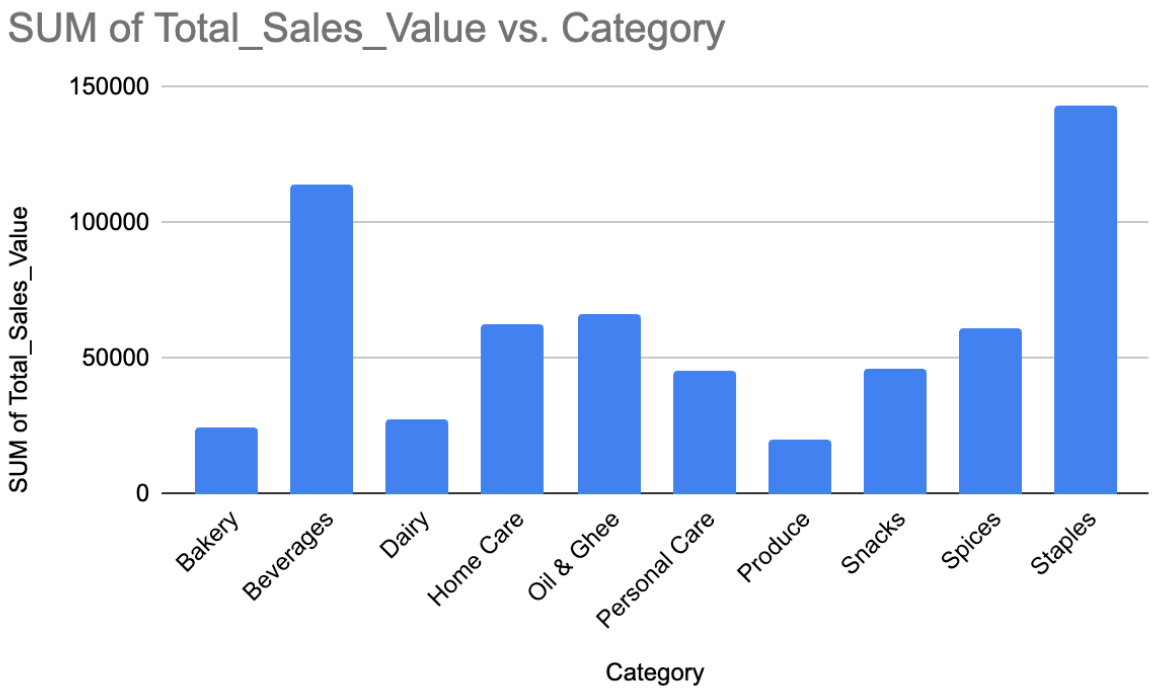
The combined use of **EDA and MBA** is the most appropriate method for understanding customer purchase behavior within a small-scale retail environment. EDA provides a descriptive layer of insight, enabling the detection of sales patterns and customer preferences, while MBA introduces a predictive layer by quantifying co-purchase relationships.

This approach was chosen for its interpretability and practical value. The Apriori algorithm is particularly well-suited to low-dimensional transactional datasets like that of Nana Grocery Store, where transactions are numerous but product variety is manageable.
Unlike more complex machine learning models, Apriori offers transparent, rule-based outcomes that are easy for non-technical business owners to understand and apply.

The overall methodological workflow — **data cleaning → exploratory analysis → association rule mining → visualization → interpretation** — ensures analytical rigor, reproducibility, and alignment with the project's problem statement. It allows the store to move from intuition-driven to **evidence-based decision-making**, improving inventory planning, cross-selling, and operational efficiency.

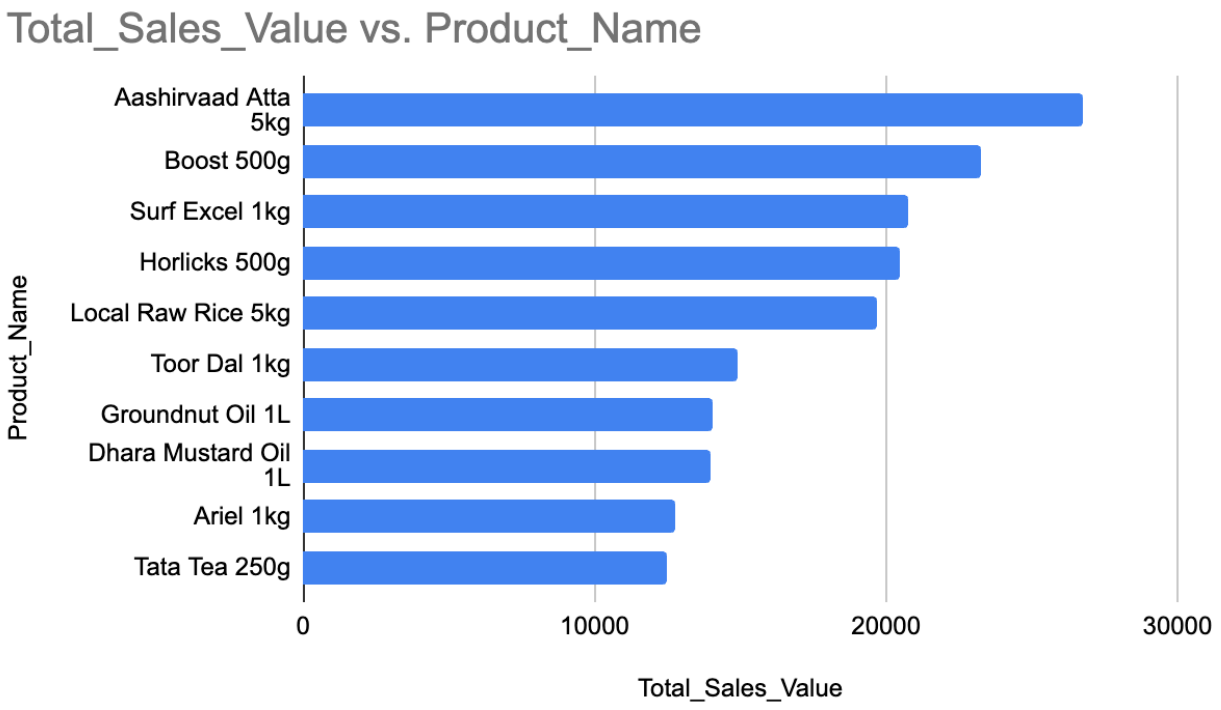
5. Results and Findings

5.1 Category-Wise Total Sales



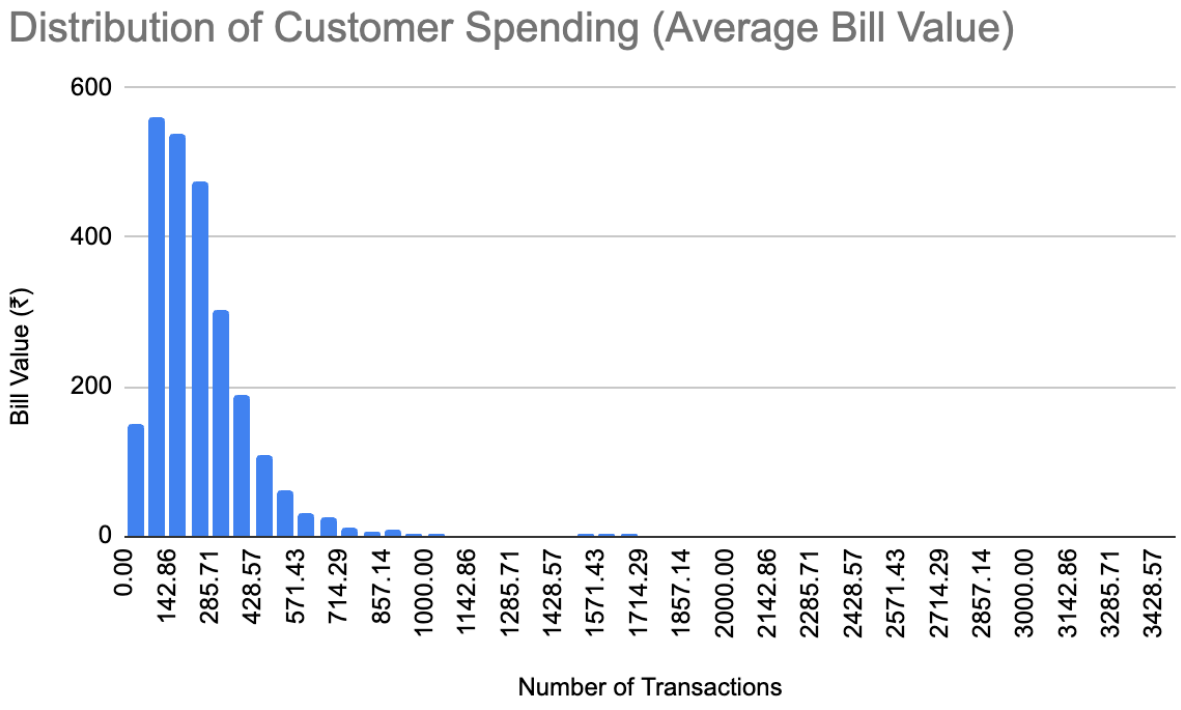
- Category Performance Overview:The bar chart displays total sales value distribution across ten product categories. Staples and Beverages emerge as the dominant revenue contributors, together accounting for the largest share of overall sales during the three-month period.
- High-Performing Categories:Staples generate the highest total sales (around ₹1.4 lakh), followed closely by Beverages (₹1.1 lakh). This pattern indicates consistent demand for essential daily-use items and high turnover of packaged drinks, aligning with regular household purchase habits.
- Moderate Contributors:Oil & Ghee, Home Care, and Personal Care categories record moderate sales levels, reflecting steady but non-seasonal buying behavior. These items are typically replenished on a biweekly or monthly basis, contributing stable revenue without significant peaks.
- Low-Contribution Categories:Produce, Bakery, and Dairy categories show comparatively lower sales volumes. This trend may stem from perishability, limited shelf life, or smaller product range, suggesting potential areas for expansion or promotional focus to increase movement.
- Business Implications:The sales distribution reveals a dependence on core household essentials. To diversify revenue, the store can introduce cross-promotions between high-volume (Staples) and moderate-volume (Snacks or Spices) categories. This will help balance sales performance and encourage multi-category purchases.

5.2 Top 10 Products by Total Sales



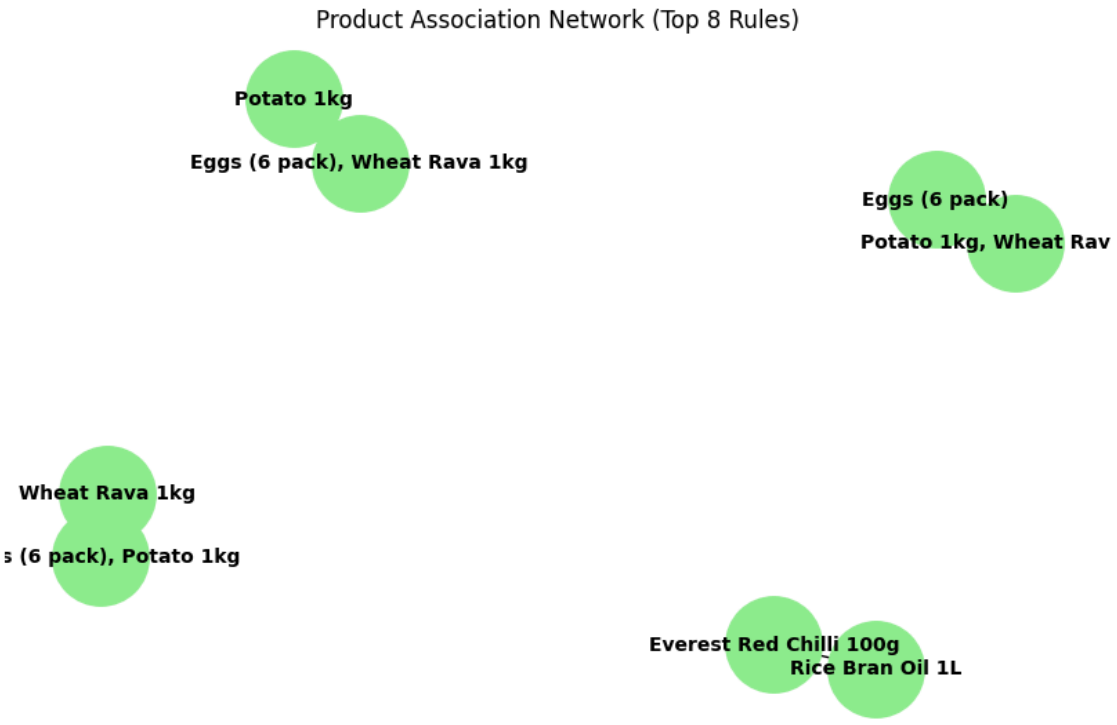
- Top-Performing Products: The chart highlights Aashirvaad Atta 5kg as the single highest-selling product, followed closely by Boost 500g, Surf Excel 1kg, and Horlicks 500g. These four products together contribute a major share of the total sales revenue, indicating strong customer loyalty and repeat purchasing.
- Core Household Staples: Products like Aashirvaad Atta, Local Raw Rice 5kg, and Toor Dal 1kg represent essential grocery staples, driving steady sales throughout the observed period. Their high ranking confirms that basic food grains remain the backbone of the store's revenue model.
- High-Value FMCG Items: The inclusion of Surf Excel 1kg and Ariel 1kg among the top sellers shows that branded detergents contribute significantly to revenue despite being higher-priced items. This suggests that customers prefer trusted household brands over low-cost alternatives.
- Health and Beverage Segment: Boost 500g, Horlicks 500g, and Tata Tea 250g reflect consistent demand in the health drinks and beverage segment. Their strong sales performance highlights the store's alignment with consumer preferences for daily consumption and nutritional products.
- Strategic Implication: The dominance of a few top-selling products suggests a concentrated sales mix. To enhance profitability, Nana Grocery Store can introduce targeted promotions for mid-tier products like Groundnut Oil and Dhara Mustard Oil, leveraging cross-selling opportunities with high-performing staples.

5.3 Average Bill Value



- Spending Range and Concentration: The histogram shows that most transactions fall within the ₹100–₹500 range, with a clear concentration of bills between ₹200 and ₹350. This indicates that the majority of customers spend modestly per visit, reflecting small, need-based purchases.
- Peak Transaction Value: The most frequent bill values occur around ₹250–₹300, suggesting that the typical transaction size for Nana Grocery Store hovers within this range. This represents the average daily household purchase pattern for grocery essentials.
- Right-Skewed Distribution: The chart displays a long tail extending beyond ₹800, confirming that while high-value transactions exist, they occur infrequently. Only a small proportion of customers make purchases exceeding ₹1000, indicating limited bulk-buying activity.
- Customer Behavior Insight: The dominance of mid-range bills implies that customers tend to visit frequently for smaller, regular purchases rather than infrequent bulk shopping. This pattern aligns with the store's neighborhood convenience model catering to local households.
- Business Opportunity: By introducing combo offers or targeted promotions — such as "Buy 2 save 10%" or "Staple + Beverage" bundles — the store could strategically increase the average bill value from ₹300 to ₹400 without needing to expand its customer base. This shift could directly improve total revenue and inventory turnover.

5.4 Product Association Rules



- The network graph depicts the top product association rules derived using the Apriori algorithm. Each node represents a product, and edges indicate the co-purchase relationships, with thickness corresponding to the strength of the lift value. The visual clustering shows distinct product groups frequently purchased together, such as *Eggs (6 pack)*, *Potato 1kg*, and *Wheat Rava 1kg*.
- A strong association exists between *Eggs (6 pack)*, *Potato 1kg*, and *Wheat Rava 1kg* (Lift ≈ 8.0), highlighting that these core household staples are often purchased together. This pattern indicates a common consumer tendency to restock essential breakfast and meal items simultaneously.
- *Rice Bran Oil 1L* and *Everest Red Chilli 100g* form another recurring pair (Lift ≈ 3.18), suggesting complementary purchase behavior in daily cooking needs. This trend reflects rational consumer habits where customers buy ingredients used together in food preparation.
- A moderate association between *Horlicks 500g* and *Real Fruit Juice 1L* (Lift ≈ 3.08) reveals a preference pattern among health-conscious consumers, likely targeting morning or evening refreshment needs. Such cross-category linkages between nutrition and beverages indicate opportunities for co-promotional offers.
- These findings demonstrate that consumers at Nana Grocery Store exhibit structured and predictable co-purchasing behavior. Promoting these high-lift pairs through shelf adjacency, combo discounts, or weekend offers could significantly improve average transaction value and encourage repeat purchases.